Valoración automática de viviendas mediante modelos predictivos



Análisis de Datos Espaciales
Facultad de Estudios Estadísticos

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01

DEPURACIÓN EXPLORACIÓN



PREPROCESAMIENTO DE DATOS

Actualización precio de la vivienda

Ministerio de Vivienda

- 2010 (T1): 3296.3 €/m²
- 2025 (T4): 5286.2 €/m²

Variación = (5286.2 − 3296.3) / 3296.3 = 0.603

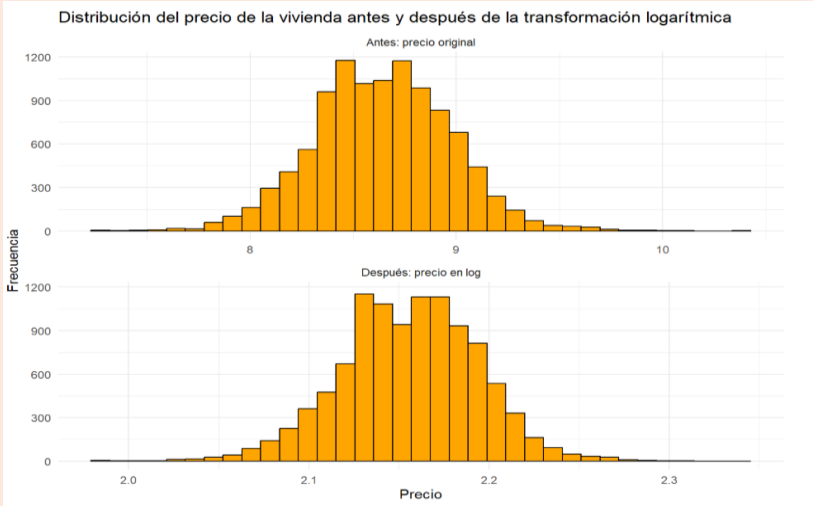
Precio actual = Precio 2010 × (1 + 0.603)



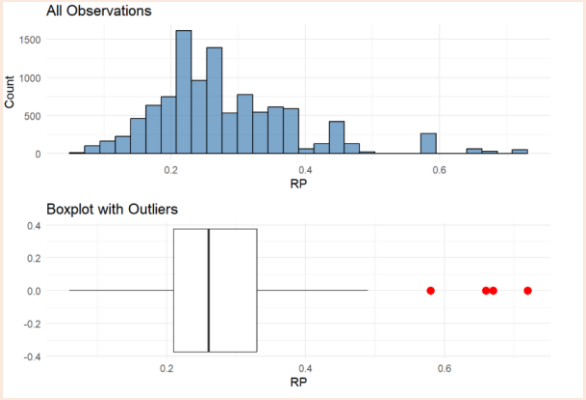
Log precio de la vivienda

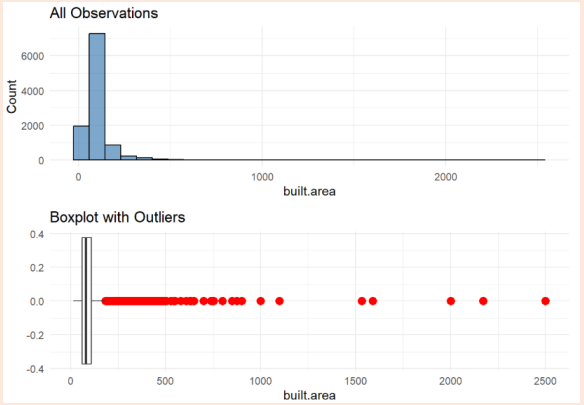
- Distribución asimétrica

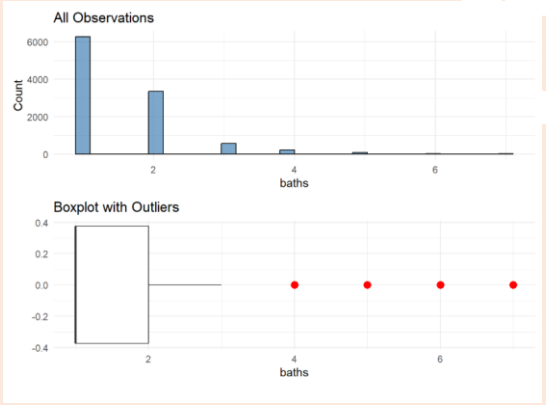
House.Price = log(house.Price)

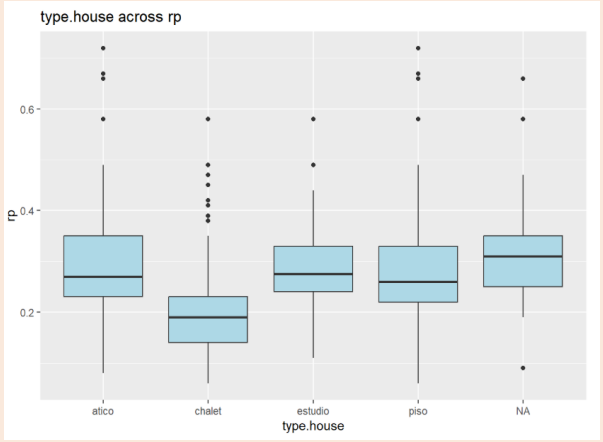


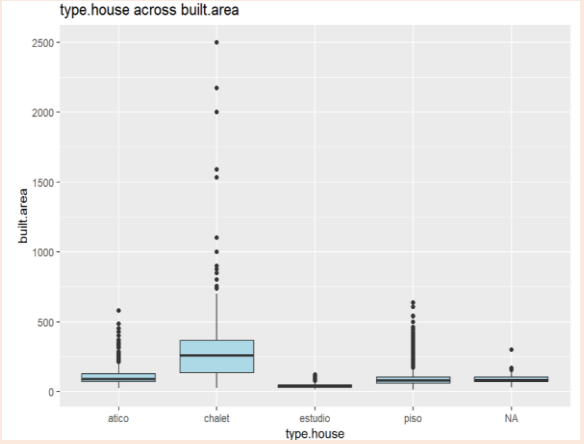
ANÁLISIS DE OUTLIERS

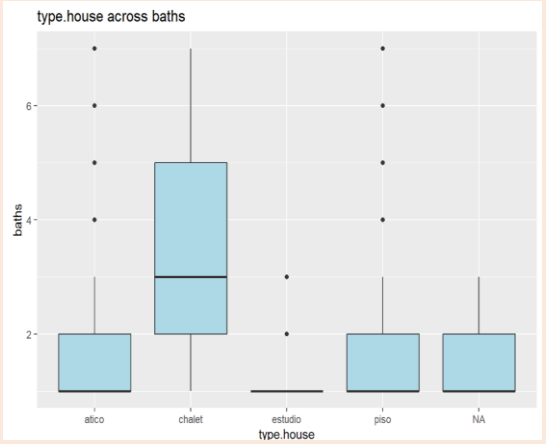










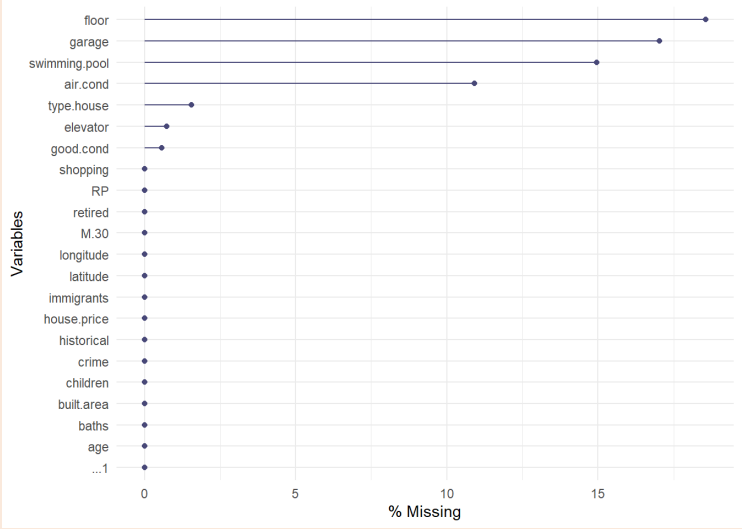


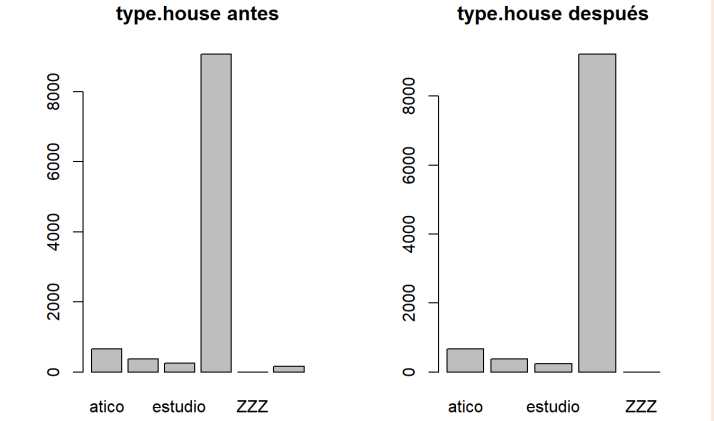
ANÁLISIS DE NAS

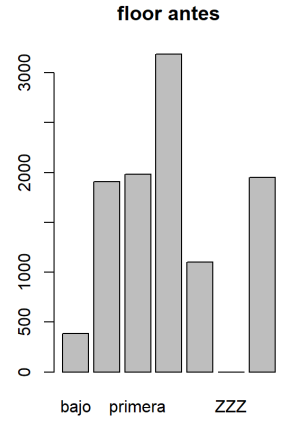
ZZZ → NA

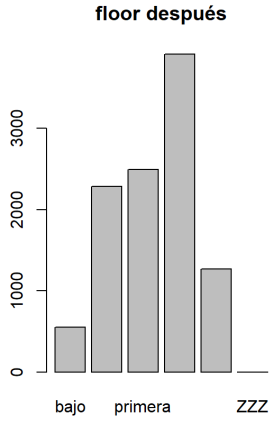
Binarias “si” o “no” → NO

Resto → Imputación MICE



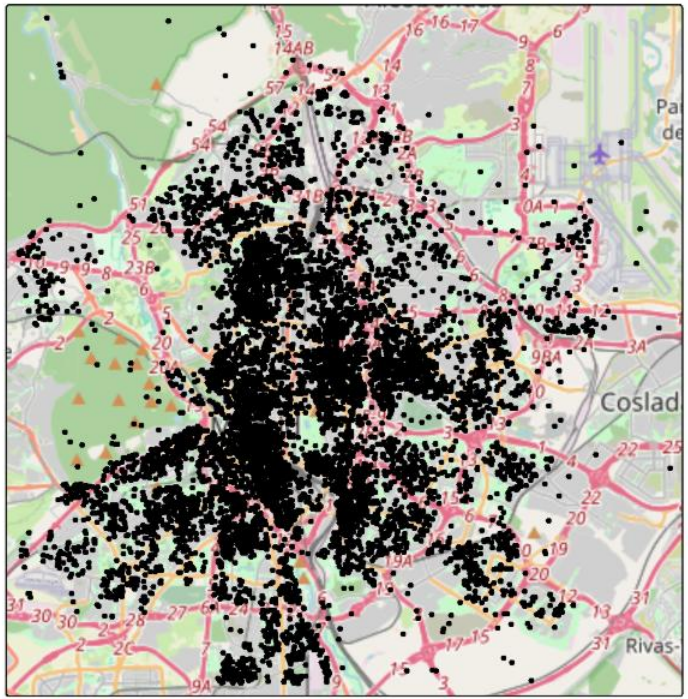








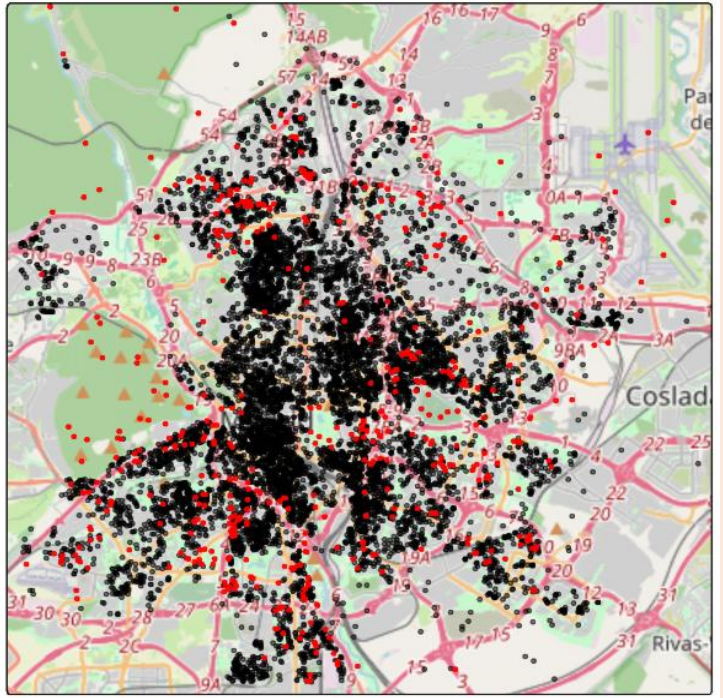
Eliminación de viviendas en zonas no residenciales



IGN USO DEL SUELO

Eliminamos zonas “no residenciales”

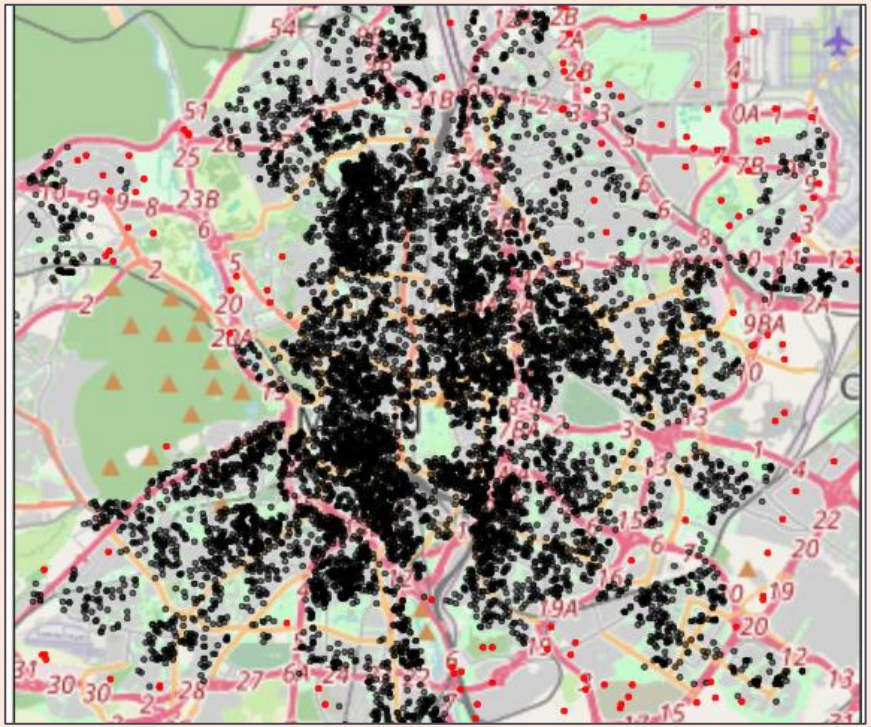
Eliminamos alguna manualmente



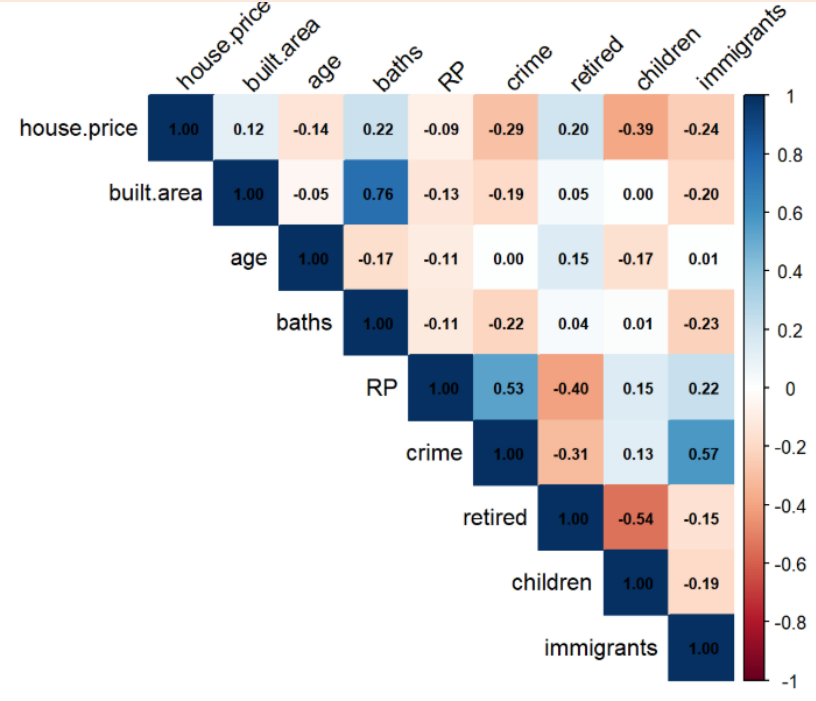
Eliminación de registros espacialmente aislados

3 vecinos o menos en un radio de 500 metros

VIVIENDA AISLADA



Exploración de datos



Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
shopping	0	1	FALSE	2	0: 5730, 1: 4098
historical	0	1	FALSE	2	0: 6284, 1: 3544
type.house	0	1	FALSE	4	pis: 8619, ati: 642, cha: 331, est: 236
floor	0	1	FALSE	5	seg: 3677, pri: 2326, cua: 2149, sex: 1169
good.cond	0	1	FALSE	2	bue: 8296, a_r: 1532
garage	0	1	FALSE	2	no: 7033, si: 2795
elevator	0	1	FALSE	2	si: 6426, no: 3402
air.cond	0	1	FALSE	2	no: 5284, si: 4544
swimming.pool	0	1	FALSE	2	no: 8145, si: 1683
M.30	0	1	FALSE	2	0: 8434, 1: 1394

Variable type: numeric

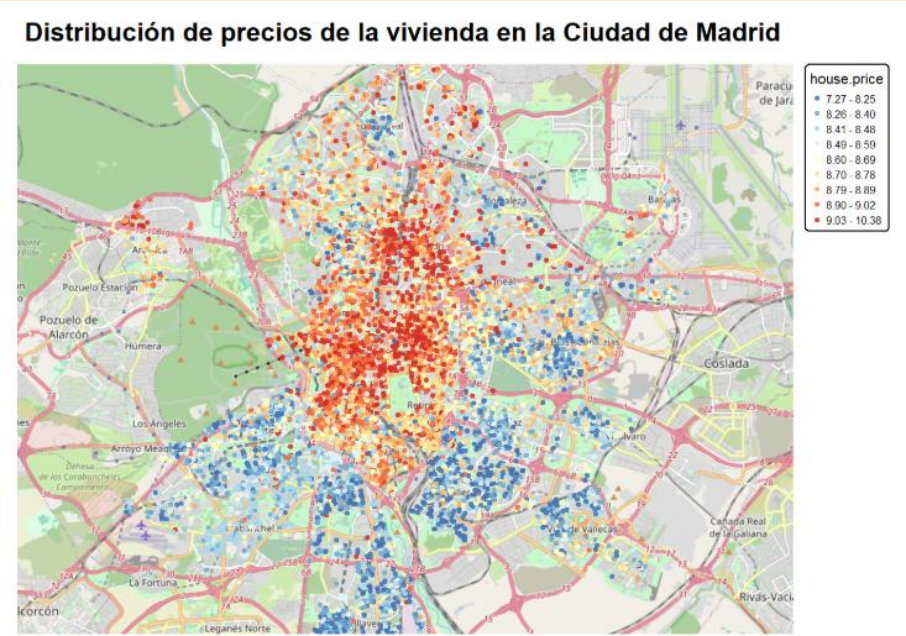
skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
house.price	0	1	8.65	0.33	7.27	8.42	8.66	8.87	10.38	
RP	0	1	0.28	0.10	0.08	0.22	0.26	0.33	0.72	
crime	0	1	0.44	0.16	0.15	0.29	0.43	0.56	0.77	
built.area	0	1	98.06	77.98	15.00	62.00	80.00	110.00	2500.00	
age	0	1	23.96	12.34	0.00	15.00	29.23	30.00	100.00	
retired	0	1	18.56	2.75	11.58	16.56	18.68	20.78	22.75	
children	0	1	12.83	2.13	8.90	11.18	12.78	14.20	18.37	
immigrants	0	1	12.68	4.64	6.06	8.04	12.07	17.56	20.54	
baths	0	1	1.54	0.81	1.00	1.00	1.00	2.00	7.00	
longitude	0	1	-3.69	0.04	-3.79	-3.71	-3.69	-3.66	-3.57	
latitude	0	1	40.42	0.04	40.34	40.40	40.42	40.45	40.51	

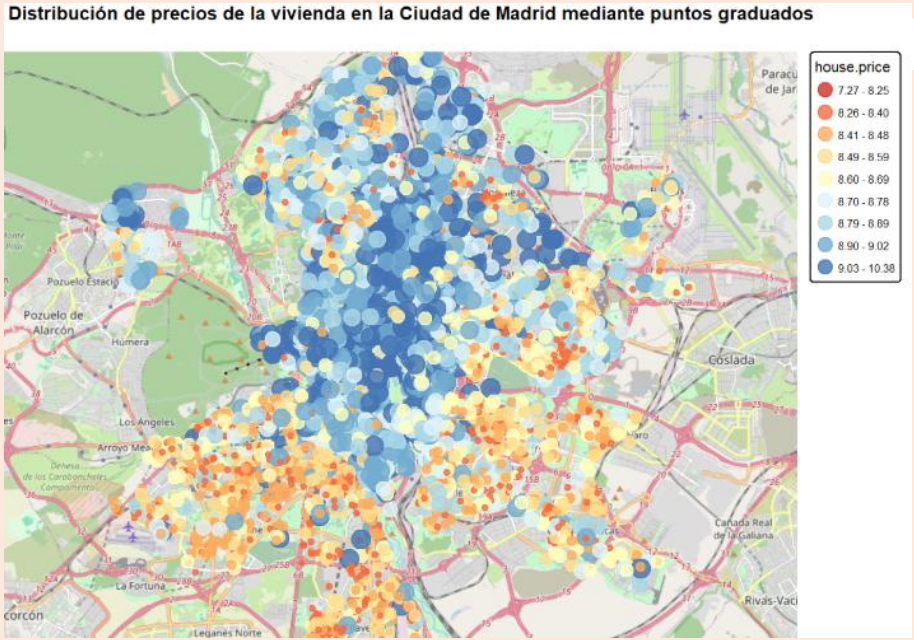
02

ANÁLISIS ESPACIAL



¿Dependencia Espacial?





Precios similares en viviendas cercanas

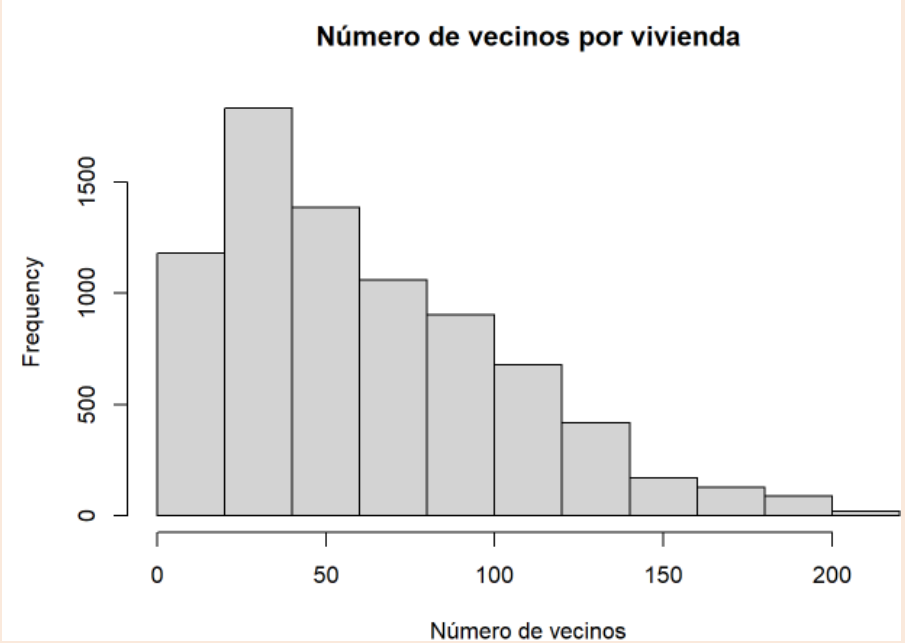
División en train y test



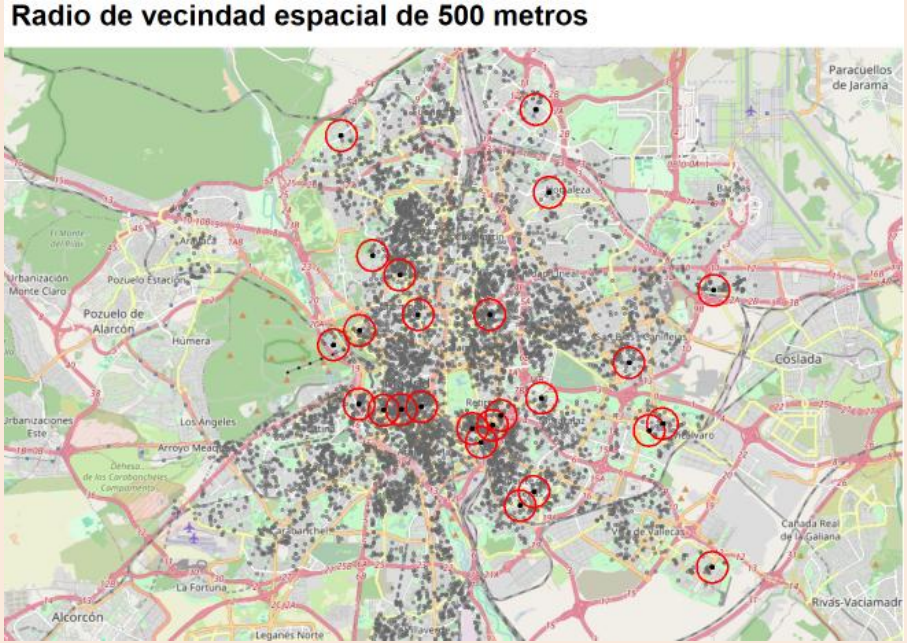
80% Train · 20% Test

Homogeneidad espacial de ambas muestras en toda la ciudad de Madrid ✅

Matriz W de pesos espaciales

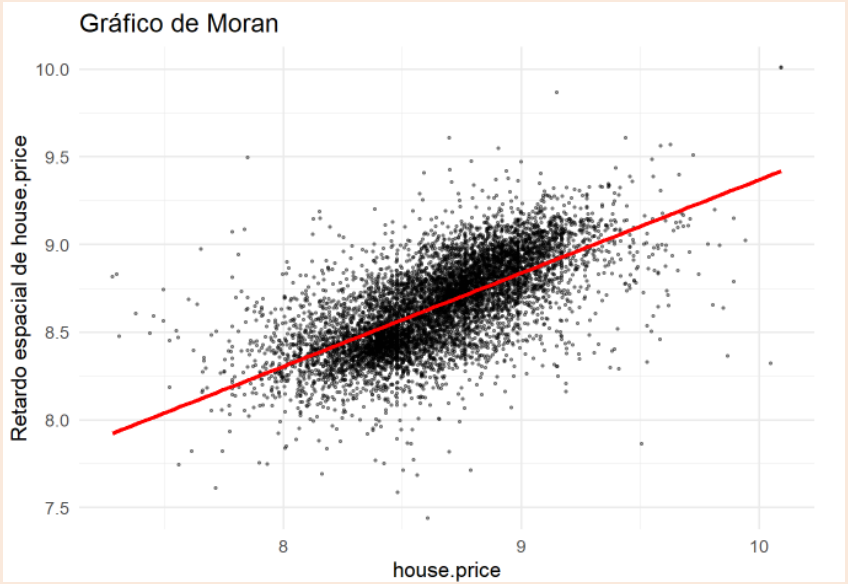


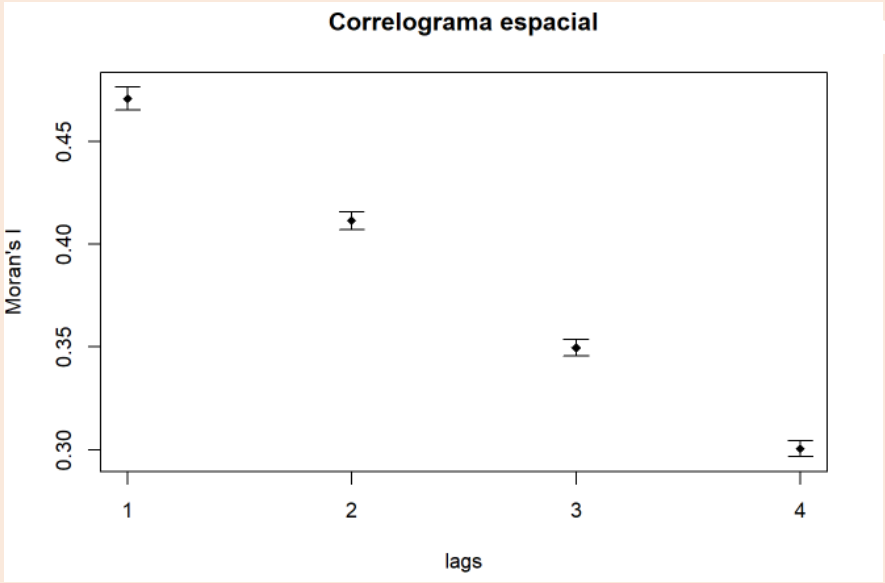
Media → 63 vecinos



Inversa del Cuadrado de la distancia entre viviendas

Autocorrelación Espacial Global ✅



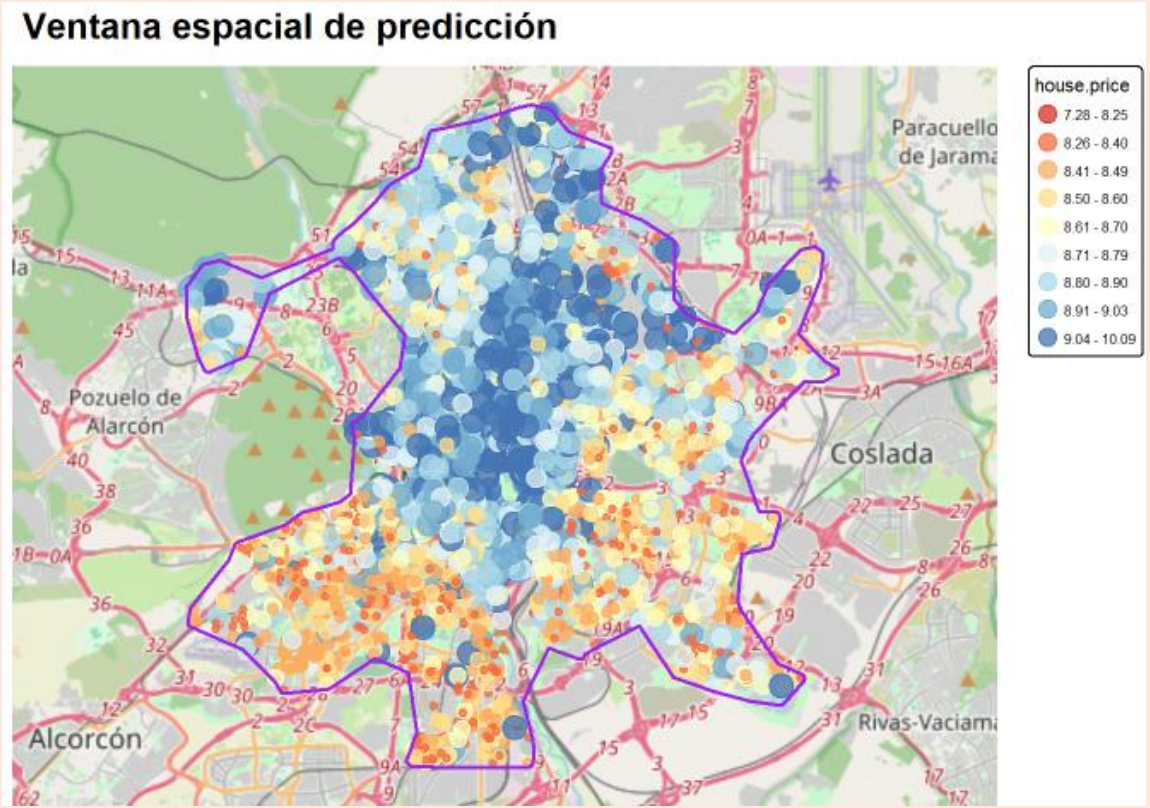


Estadístico I de Moran = 0.5318

Disminución progresiva

Ventana Espacial de Predicción

- Delimita donde existe suficiente densidad de viviendas de **entrenamiento**
- Si la vivienda queda fuera, predicción **no fiable**



03

MODELIZACIÓN



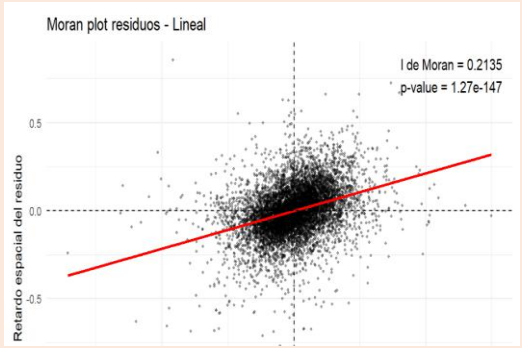
MODELOS ML SIN ESPACIALIDAD

Modelo	RMSE	MAE	R^2
Random Forest	0.200	0.141	0.647
XGBoost	0.213	0.148	0.603
SVR	0.213	0.151	0.599
Red Neuronal	0.226	0.164	0.545
Lineal	0.236	0.174	0.506
KNN	0.241	0.172	0.494

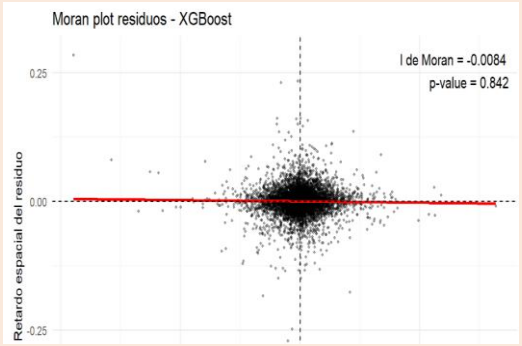


¿Dependencia espacial en los residuos?

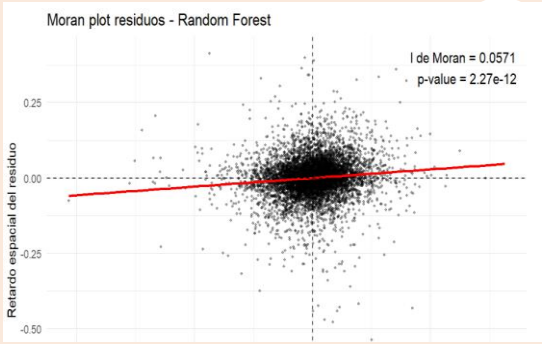
Lineal



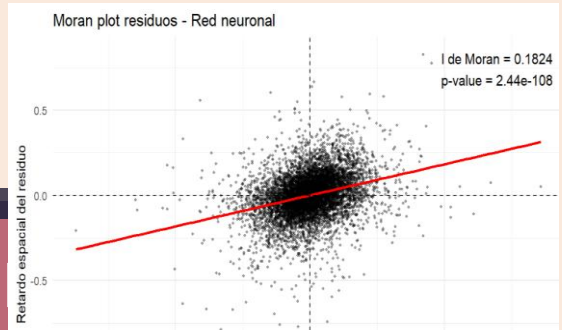
XGboost



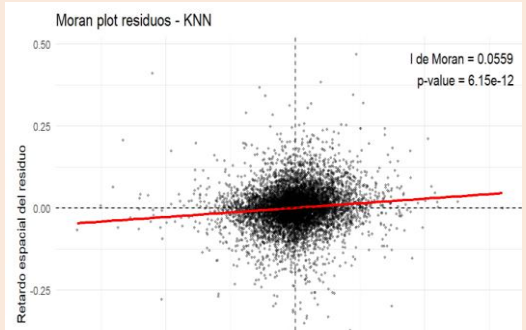
Random Forest



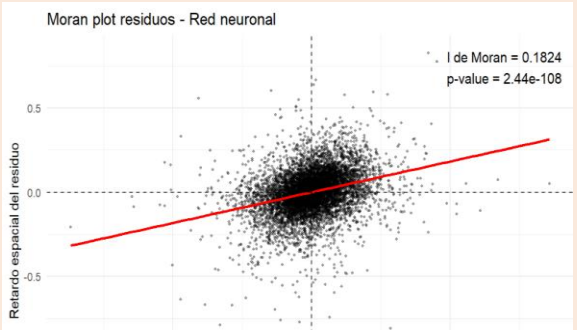
SVM



KNN



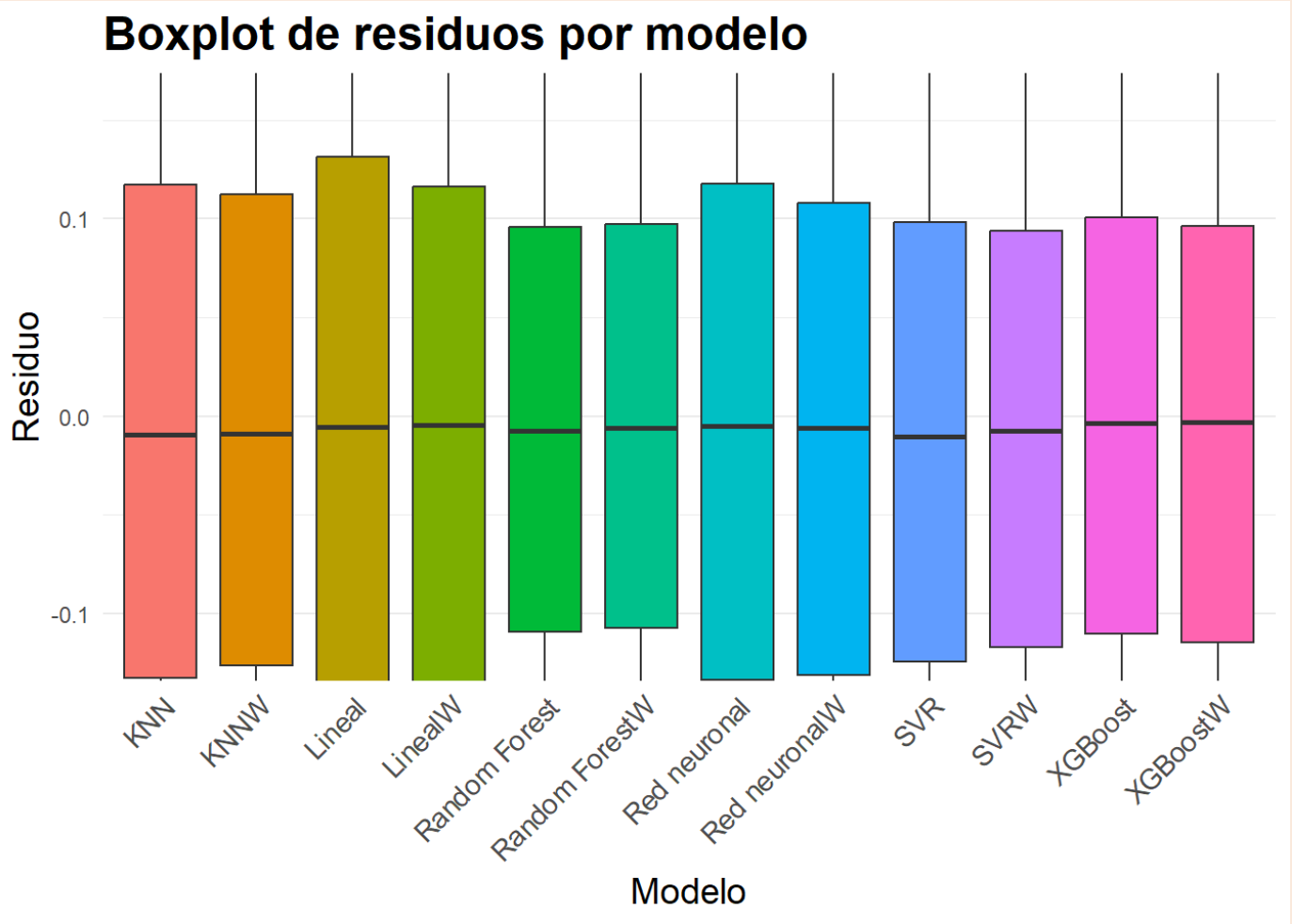
Red Neuronal



MODELOS ML INCORPORANDO ESPACIALIDAD

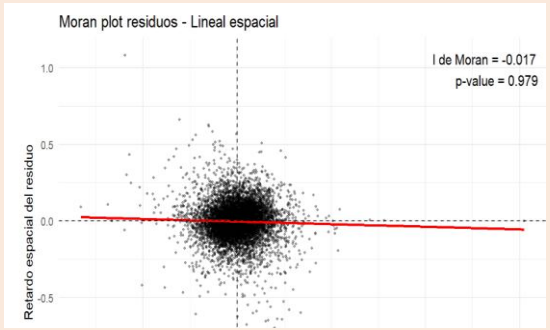
MODELO	RMSE	MAE	R^2
Random Forest espacial	**0.200**	**0.140**	**0.646**
Random Forest	0.200	0.141	0.647
XGBoost espacial	**0.206**	**0.146**	**0.628**
SVR espacial	**0.209**	**0.147**	**0.613**
XGBoost	0.213	0.148	0.603
SVR	0.213	0.151	0.599
Red neuronal espacial	**0.221**	**0.159**	**0.568**
Lineal espacial	**0.225**	**0.163**	**0.549**
Red neuronal	0.226	0.164	0.545
KNN espacial	**0.237**	**0.168**	**0.508**
Lineal	0.236	0.174	0.506
KNN	0.241	0.172	0.494



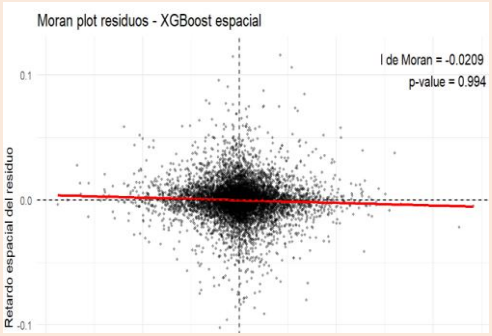

Boxplot de residuos por modelo
Residuo
0.1
0.0
-0.1
KNN
KNNW
Lineal
LinealW
Random Forest
Random ForestW
Red neuronal
Red neuronalW
SVR
SVRW
XGBoost
XGBoostW
Modelo

¿Dependencia espacial en los residuos?

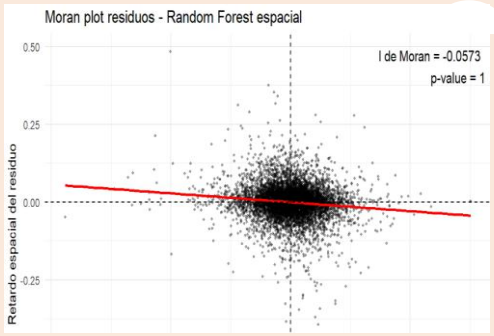
LinealW



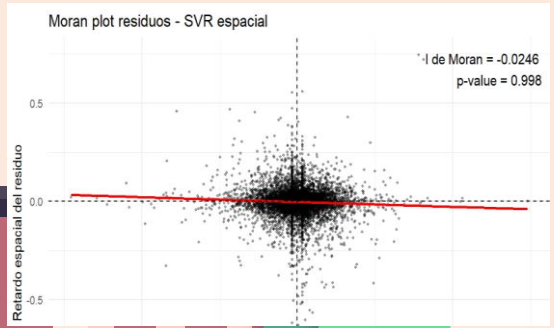
XGboostW



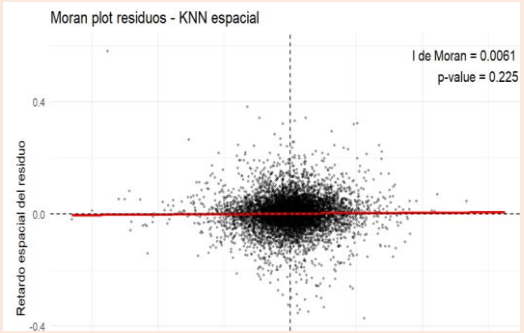
Random ForestW



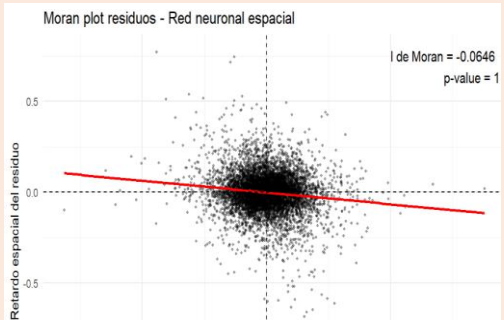
SVMW



KNNW



Red NeuronalW



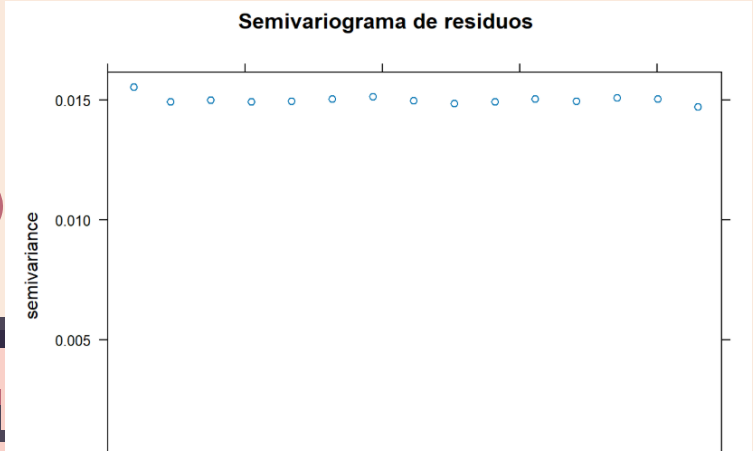
MODELO KRIGING

Kriging sobre los residuos

$$\hat{y}(s_i) = \hat{f}(x_i) + \hat{u}(s_i)$$

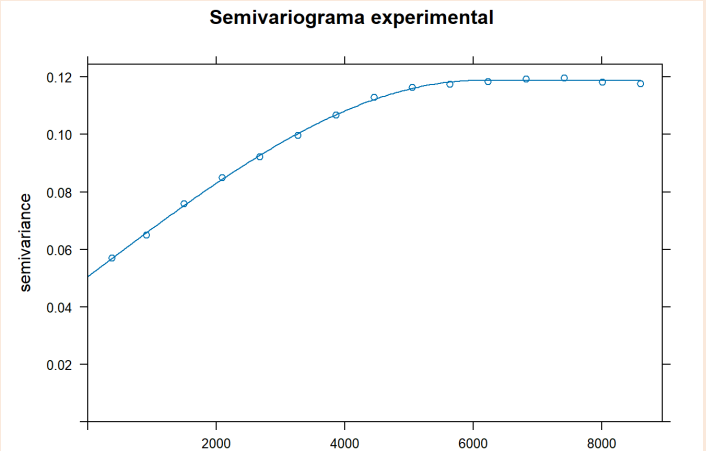
$\hat{f}(x_i)$ → predicción obtenida mediante el modelo de Machine Learning

$\hat{u}(s_i)$ → componente espacial estimada mediante kriging sobre los residuos



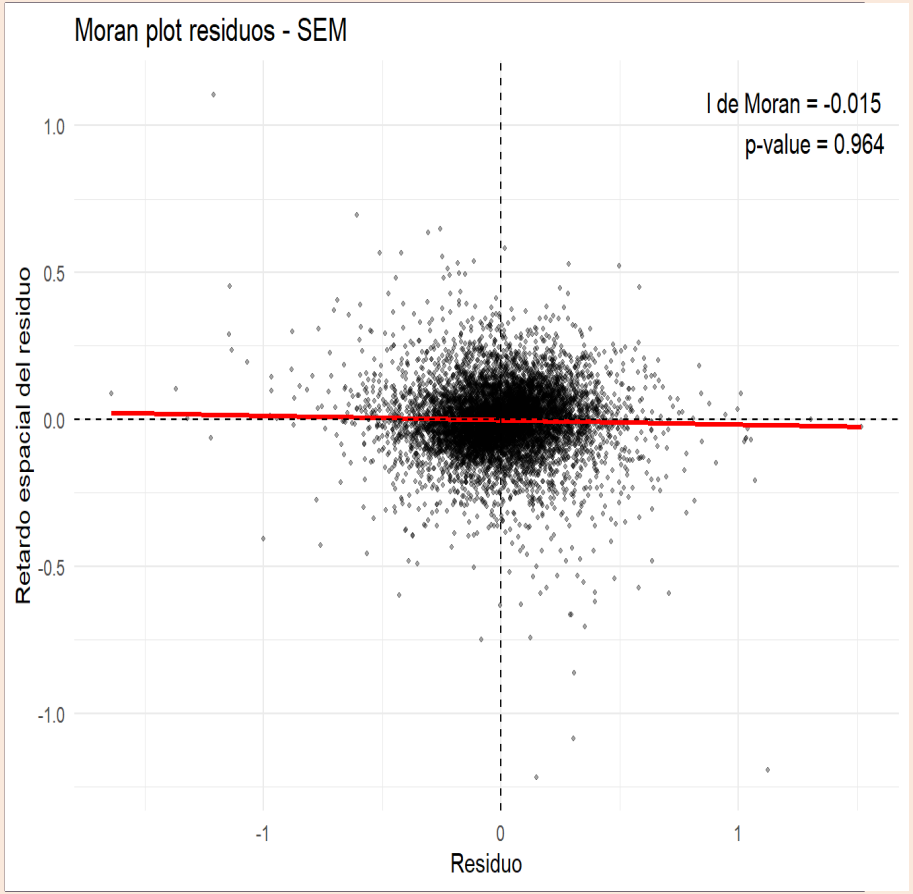
Kriging Ordinario

- Utiliza únicamente la dependencia espacial entre viviendas cercanas para realizar predicciones
- Permite evaluar la capacidad predictiva de la información espacial por sí sola



MODELO ESPACIAL SEM

- Introduce la dependencia espacial en los errores del modelo
- Asume que viviendas cercanas comparten información no observada
- Corrige patrones espaciales residuales no explicados por las variables predictoras
- No modifica directamente la variable objetivo, sino el término de error
- Pertenece a los modelos clásicos de econometría espacial



04

RESULTADOS

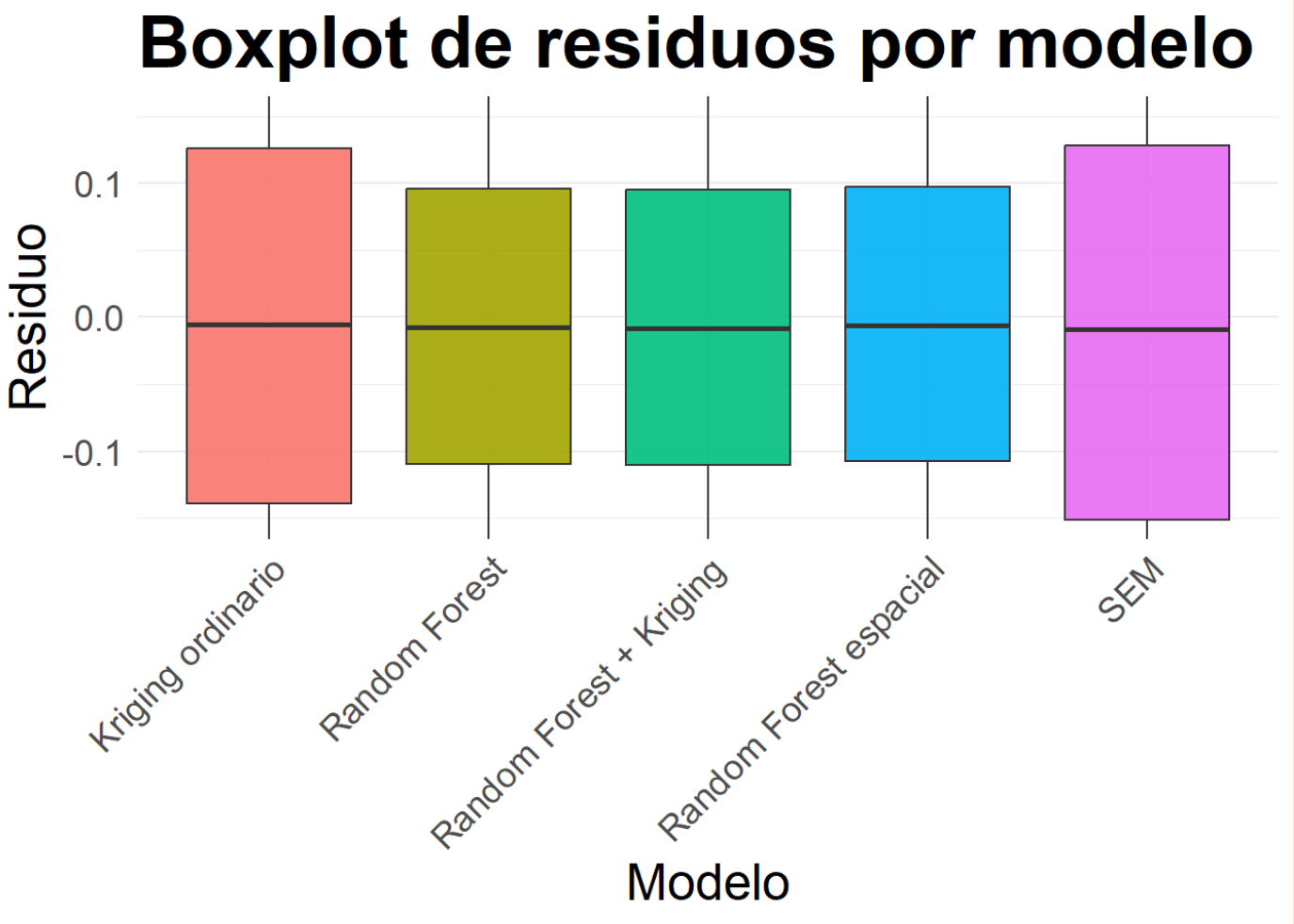


COMPARACIÓN DE TODOS LOS MODELOS

MODELO	RMSE	MAE	R^2	TIPO
Random Forest espacial	**0.200**	**0.140**	**0.646**	ML + retardo espacial
RF + kriging residuos	**0.200**	**0.141**	**0.647**	Híbrido
Random Forest	0.200	0.141	0.647	ML
XGBoost espacial	**0.206**	**0.146**	**0.628**	ML + retardo espacial
SVR espacial	**0.209**	**0.147**	**0.613**	ML + retardo espacial
XGBoost	0.213	0.148	0.603	ML
SVR	0.213	0.151	0.599	ML
Red neuronal espacial	**0.221**	**0.159**	**0.568**	ML + retardo espacial
Lineal espacial	**0.225**	**0.163**	**0.549**	ML + retardo espacial
Red neuronal	0.226	0.164	0.545	ML
KNN espacial	**0.237**	**0.168**	**0.508**	ML + retardo espacial
KNN	0.241	0.172	0.494	ML
Kriging ordinario	0.238	0.173	0.497	Geoestadístico
Lineal	0.236	0.174	0.506	ML
SEM	0.237	0.175	0.502	Modelo espacial clásico

- **La espacialidad por sí sola no es suficiente**
- **ML + ESPACIAL** ✅




Boxplot de residuos por modelo
Residuo
0.1
0.0
-0.1
Kriging ordinario
Random Forest
Random Forest + Kriging
Random Forest espacial
SEM
Modelo

05

PUESTA EN PRODUCCIÓN



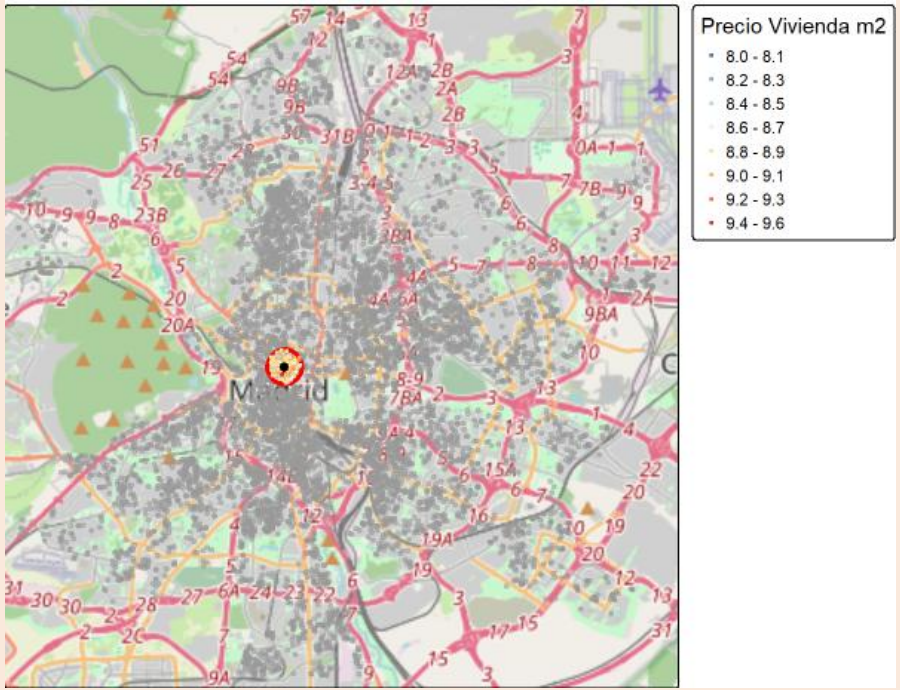
Predicción Puntual -> Nuevo Dato

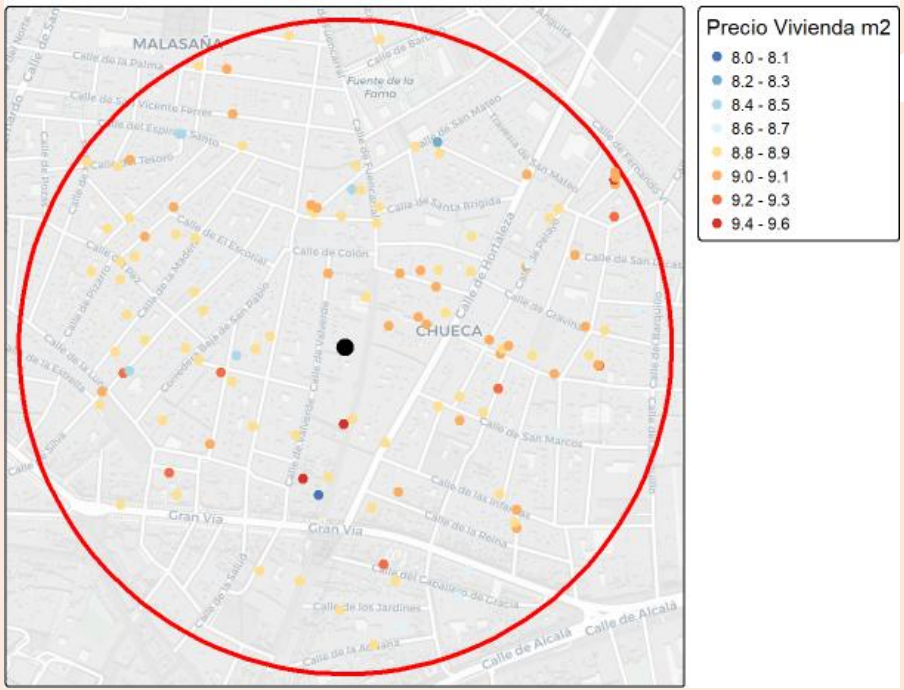
Característica	Valor
RP	0.20
Crime	0.35
Superficie construida (m²)	80
Antigüedad	30
Jubilados (%)	18
Niños (%)	12
Inmigrantes (%)	10
Zona comercial	No
Zona histórica	No
Baños	2
Tipo de vivienda	Piso
Planta	Segunda - tercera
Estado	Bueno
Garaje	No
Ascensor	Sí
Aire acondicionado	Sí
Piscina	No
Dentro M-30	Sí
Longitud	-3.7012
Latitud	40.4225

"La vivienda está dentro de la ventana de observación"



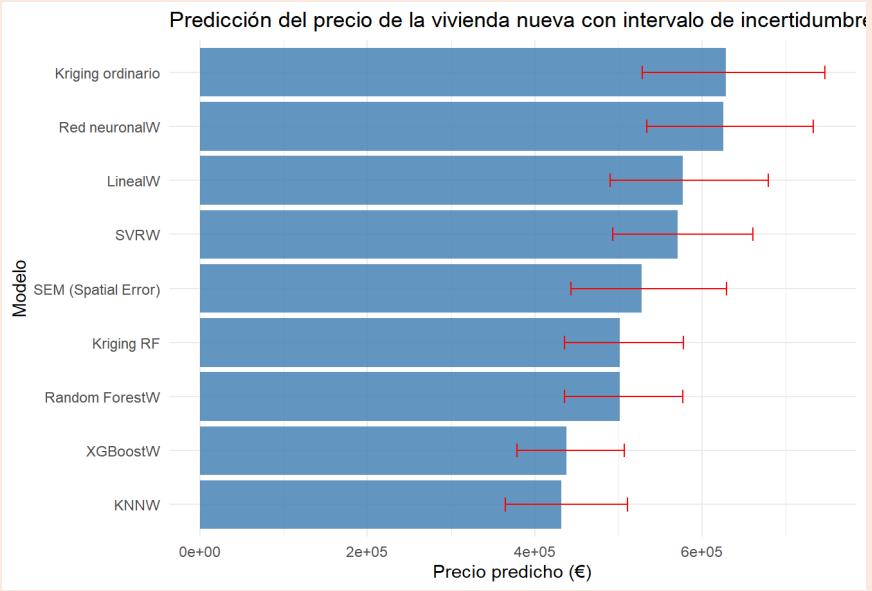
Representación de la nueva vivienda





PREDICCIÓN E INTERVALOS

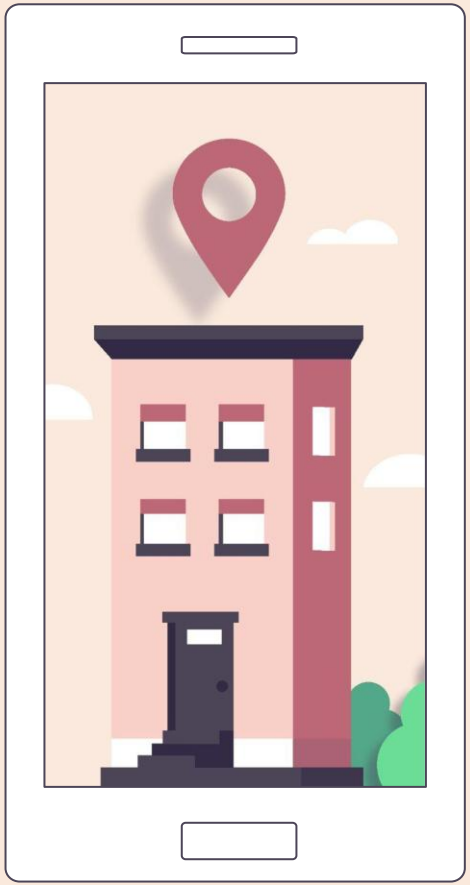
Modelo	MAE	Precio vivienda (€)	Intervalo inferior (€)	Intervalo superior (€)
Random Forest espacial	0.140	501443	435909	566977
Kriging RF	0.141	501673	435634	567712
XGBoost espacial	0.146	438161	378486	497836
SVR espacial	0.147	570749	492941	648557
Red neuronal espacial	0.159	625652	533897	717407
Lineal espacial	0.163	577080	490395	663765
KNN espacial	0.168	431625	364908	498342
Kriging ordinario	0.173	628212	528327	728097
SEM	0.175	527835	442911	612759
IDW	NA	633345	NA	NA
Media ponderada	NA	527489	NA	NA



06

APLICACIÓN SHINY





https://geovaluemadrid.shinyapps.io/geovalue_madrid/

¡GRACIAS!

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Mario Gómez Fuentes · Marcos Martínez Aguiar

